AI NOTES SERIES

AI-Assisted Data Analysis

Simple, clear notes on turning everyday records into answers - collecting and cleaning data, using basic statistics and charts, and letting AI help while you stay in charge of the checking. For teachers, students, and beginners.

By Shah Fahad

AI & Technology Innovator and Educator · Buner, Khyber Pakhtunkhwa, Pakistan

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THE BASICS

What Data Analysis Is

Data analysis means turning raw records into clear answers. Raw records on their own - a long list of marks, a register full of ticks - do not tell us very much. Analysis is the work of organising those records, checking them, and summarising them so that a real question can be answered: Which topic did the class find hardest? Is attendance better this month than last month? Where should we give extra help?

Data as rows and columns. Most data is kept in a table. Each **row** is one record - one student, one day, one sale. Each **column** is one kind of information - a name, a date, marks in English, marks in mathematics. When data is arranged in neat rows and columns, a person - or a computer - can sort it, count it, and compare it quickly and correctly.

Examples from school life

- **Attendance:** the daily register becomes a simple count - how many students were present each day, and who is absent often.
- **Marks:** test scores become averages and comparisons - which subject is strong, and which students need support.
- **Results over time:** term-by-term results show whether the class is improving, staying the same, or slipping.

Examples from daily life

- A shopkeeper counts daily sales to learn which items sell best and when to order more stock.
- A family notes monthly electricity bills to see in which months the solar system or the grid carries the bigger load.
- A health worker counts patients each week to spot whether an illness is spreading in an area.

> **A simple way to remember it:** data is the raw material, analysis is the careful work, and the answer - a number, a chart, or a short written finding - is the product. Good analysis always starts with a clear question, not with the numbers.

These notes come from practical work with school data, student results, and research data - the kind of records a teacher handles every week. No advanced mathematics is needed. Careful checking matters more than clever methods.

THE METHOD

The Process, Step by Step in Words

Reliable analysis follows the same order every time. Skipping a stage - especially cleaning - is the most common reason an answer turns out wrong.

Collect → **Clean** → **Explore** → **Use simple statistics** → **Visualise** → **Interpret** → **Report** → **Quality-check**

Collect. Gather the records you need in one place - the register, the marks sheet, the survey forms. Write down where each record came from and what period it covers, so the source can be checked later.

Clean. Fix the mistakes before any counting begins. Correct typing errors, remove duplicate rows where the same record was entered twice, and mark missing values honestly instead of guessing them. A clean table is worth more than a fast calculation on a dirty one.

Explore. Look at the data before summarising it. Read down the columns, sort from smallest to largest, and notice anything unusual - a mark of 105 out of 100, or one day with no attendance recorded at all.

Use simple statistics. Summarise with a few honest numbers - totals, averages, and percentages. The next chapter explains the four summaries you will use most.

Visualise. Turn the summary into a chart. A good chart lets a reader see the pattern in seconds, without reading every row of the table.

Interpret. Say what the numbers mean, in plain words, and only what they actually show. A rise in marks after extra classes suggests the classes helped - it does not prove it on its own.

Report. Share the question, the key numbers, the chart, and the conclusion together, so any reader can follow how the answer was reached.

Quality-check. Before trusting the result, recheck a few figures by hand against the original records. If a hand check and the computer disagree, find the reason before reporting anything.

CORE SKILLS

Simple Statistics You Must Know

Four summaries answer most everyday questions. Each one below uses the same small example: the test marks of five students, out of 100.

Student	Ali	Sara	Bilal	Hina	Omar
Marks	60	70	80	80	90

Mean - the ordinary average

The mean adds all the values together and divides by how many values there are. It gives one number that stands for the whole group.

> **Worked example:** 60 + 70 + 80 + 80 + 90 = 380. There are 5 students, so the mean is 380 ÷ 5 = **76 marks**.

Median - the middle value

The median is the value in the middle when all values are arranged in order - half below it, half above it. It is useful when one very high or very low value would pull the mean away from the typical case.

> **Worked example:** in order, the marks are 60, 70, **80**, 80, 90. The middle value is **80**, so the median is 80 marks.

Mode - the most common value

The mode is the value that appears most often. It answers questions like: which mark did most students get? A set of values can have one mode, more than one, or none at all if every value appears the same number of times.

> **Worked example:** the mark **80** appears twice, and every other mark appears once. The mode is 80 marks.

Percentage - a share out of a hundred

A percentage expresses a part as a share of the whole, scaled to a hundred. It makes groups of different sizes easy to compare fairly.

> **Worked example:** on one day, 45 students were present out of 50 on the register. 45 ÷ 50 × 100 = **90%** present.

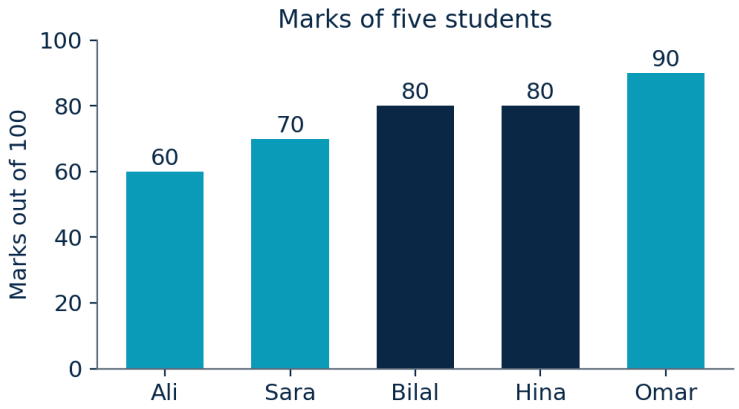
SHOWING RESULTS

Charts and What They Show

A chart turns numbers into a picture the eye can read at once. Choosing the right chart matters: each kind answers a different kind of question.

Bar chart - for comparing amounts

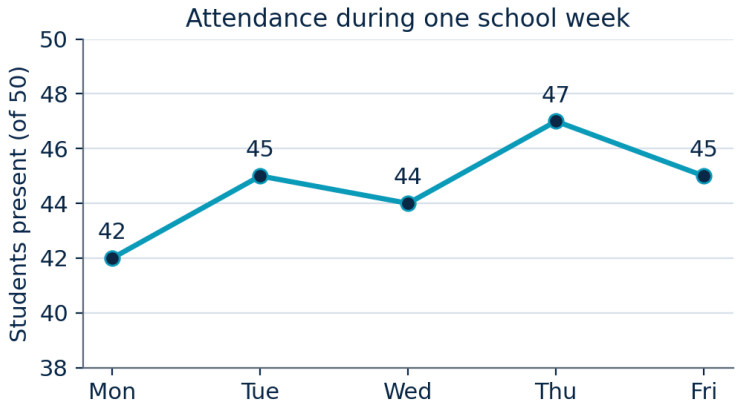
A bar chart uses bars of different heights to compare amounts between people, groups, or subjects. The taller the bar, the bigger the amount. Use it to answer: who scored more, or which subject is stronger?



The same five students from the statistics chapter, compared at a glance.

Line chart - for change over time

A line chart joins points with a line to show how something rises, falls, or stays steady over days, weeks, or terms. Use it to answer: is attendance improving, and on which days does it dip?



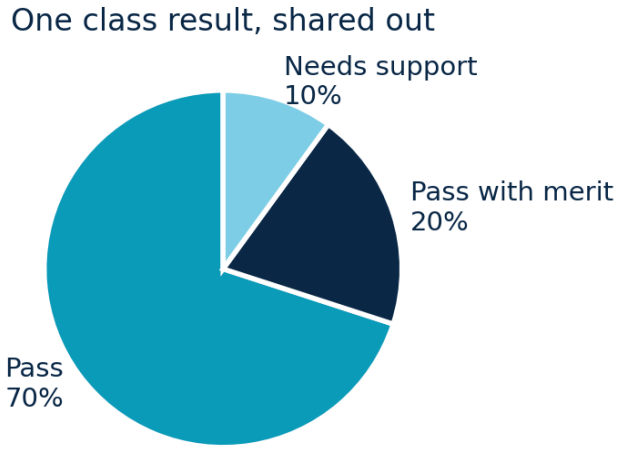
Attendance across one week - the dip on Monday is visible in a second.

SHOWING RESULTS

Charts and What They Show, Continued

Pie chart - for parts of a whole

A pie chart is a circle cut into slices. Each slice shows one part's share of the whole, and all the slices together must make exactly the whole - 100%. Use it to answer: what share of the class passed, passed with merit, or still needs support? A pie chart is a poor choice for comparing similar amounts or for showing change over time; use a bar or line chart there instead.



A simple example: the three shares add up to 100%, the whole class result.

> **Honest charts:** start a bar chart's scale at zero, label every axis and slice clearly, and never stretch a chart to make a small difference look big. A chart that misleads is worse than no chart at all.

How AI Helps

Used well, an AI assistant removes the slow, mechanical parts of analysis, so the human can spend time on judgement. In practical work with school data, results, and research data, AI helps most in these ways:

- **Writing code:** AI can draft the spreadsheet formula or the short program that sorts, counts, or averages a table, and explain each line in plain words.
- **Checking code:** AI can read a formula or program back to you and point out likely mistakes - a wrong column, a range that misses the last rows.
- **Suggesting checks:** AI can propose sensible quality checks for a dataset - looking for duplicates, impossible values, and missing entries - before you calculate anything.
- **Finding patterns in large tables:** a table with thousands of rows is too big to read by eye. AI-assisted tools can group, rank, and summarise it quickly, so patterns worth a closer look stand out.
- **Drafting explanations:** AI can turn a finished, verified result into a clear first draft of the written explanation, which the human then corrects and approves.

The Human's Job

AI works only with what it is given, and it does not know your school, your register, or your students. Some things only the human can do:

- AI cannot know that a column was recorded wrongly - for example, that absent students were marked present for a whole week by mistake.
- AI cannot know that a sample is unfair - for example, that only the strongest class was tested, so the average flatters the whole school.
- AI will calculate confidently from bad data. **Garbage in, garbage out:** wrong records in means wrong answers out, however polished they look.

> **The rule:** let AI do the counting, drafting, and suggesting. Keep the checking, the context, and the final decision for yourself - and always verify a few figures against the original records before you report them.

PRACTICE

Tools

Spreadsheets. For most school and office work, a spreadsheet program - such as Microsoft Excel or Google Sheets - is enough. It holds data in rows and columns, sorts and filters it, calculates averages and percentages with simple formulas, and draws basic charts. Learn the spreadsheet first; it covers the great majority of everyday analysis.

Python for bigger jobs. When tables grow large, or the same analysis must be repeated every month, the Python programming language is the common next step. Three of its well-known tools are named here simply and honestly:

- **Pandas** - for holding tables in rows and columns and cleaning, grouping, and summarising them.
- **NumPy** - for fast numerical calculations underneath the other tools.
- **Matplotlib** - for drawing charts, including the charts in these notes.

None of these tools thinks for you. Each one only does the counting and drawing faster; the cleaning, checking, and interpreting stay with the human, whichever tool is used.

Cautions

Correlation is not causation. When two things rise together, it does not prove that one caused the other. Ice cream sales and hot days rise together, but ice cream does not cause the heat - the summer causes both. Treat a shared pattern as a question to investigate, not as proof.

Small or biased samples mislead. Asking five students and reporting the answer as the view of a whole school is unsafe. A small sample jumps around by chance, and a biased sample - only one class, only one village, only volunteers - quietly favours one answer. Say how many records your result rests on.

Protect people's data. Marks, attendance, and health records belong to real people, many of them children. Share summaries, not named records; keep files where only authorised people can open them; and never paste private records into a public tool without permission.

REFERENCE

Words to Know

Data	Recorded facts or measurements - names, dates, marks, counts - before they are organised or explained.
Dataset	A complete collection of related data, usually one table of rows and columns kept together for one purpose.
Variable	One kind of information recorded for each row - one column, such as age, gender, or marks in English.
Mean	The ordinary average: all values added together, divided by how many values there are.
Median	The middle value when all values are arranged in order from smallest to largest.
Mode	The value that appears most often in a set of values.
Outlier	A value far away from the rest - for example, a mark of 5 among marks of 70 to 90. Outliers deserve a check: they may be real, or a recording mistake.
Correlation	A pattern where two things tend to rise or fall together. A correlation does not prove that one causes the other.
Sample	A part of a larger group, studied in order to learn about the whole group - such as one class standing for a whole school.
Bias	A tilt in the data or the sample that quietly favours one answer, such as asking only students who stayed after school.
Visualisation	Showing data as a picture - a chart or graph - so patterns can be seen quickly.

QUICK REVISION

The Whole Topic on One Page

- Data analysis turns raw records into answers; it always starts with a clear question.
- Data lives in rows (one record each) and columns (one kind of information each).
- The process: collect, clean, explore, use simple statistics, visualise, interpret, report, quality-check.
- Cleaning means fixing errors, removing duplicates, and marking missing values honestly.
- Mean is the average, median is the middle value, mode is the most common value, and percentage is a share out of a hundred.
- Bar charts compare, line charts show change over time, pie charts show parts of a whole.
- AI helps with code, checks, patterns in large tables, and drafting explanations.
- The human checks the data, knows the context, and makes the final decision - garbage in, garbage out.
- Correlation is not causation; small or biased samples mislead; protect people's privacy.

Closing note

These are introductory study notes. The explanations are simplified on purpose, so that a beginner can build a correct foundation first and add detail later. They are drawn from practical experience with school data, student results, and research data, and every worked example in them can be checked by hand.

Prepared by Shah Fahad